MANIPAL SCHOOL OF COMMERCE AND ECONOMICS

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PROJECT REPORT

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Digital Entertainment Consumption Patterns and Their Impact on Academic Engagement and Lifestyle Balance Among University Students

SUBMITTED BY

Sachin J
Shriya Kamble
Shravya Shetty
Pratheek P Rao
Thamatam Vishwa

SUBMITTED TO

Prof. Rita Rani Chopra

Faculty, Applied Multivariate Data Analysis

Manipal School of Commerce and Economics

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ABSTRACT

This study looks at how university students consume digital entertainment and how that affects their lifestyle balance and academic engagement. Students are being exposed to more digital content due to the growth of social media, gaming, and streaming platforms, which may have an impact on their daily routines and academic performance. The study examines the connection between digital entertainment time and factors like study hours, academic engagement, and sleep patterns using survey-based data. To comprehend these relationships, regression and correlation analyses are used. The results show that increased digital entertainment consumption is linked to lower academic engagement and a disturbed lifestyle balance, especially with regard to time management and sleep. On the other hand, moderate use can help reduce stress and promote relaxation. The study comes to the conclusion that although digital entertainment is an essential part of student life, excessive use can negatively affect academic outcomes and well-being.

INTRODUCTION

In recent years, the rapid advancement of digital technology has dramatically altered how people consume entertainment. Traditional forms of entertainment, such as television and physical media, have gradually given way to digital platforms such as social media, online streaming services, and interactive gaming applications. Because of their easy access to smartphones, laptops, and high-speed internet, university students are among the most active groups in terms of digital entertainment. As a result, digital entertainment has become an essential part of their daily routine.

The increased availability and accessibility of digital content has influenced students' behavior, routines, and priorities. Platforms like YouTube, Netflix, Instagram, and online gaming environments offer instant access to a wide range of content, allowing students to spend extended periods of time engaging with them. While digital entertainment can provide relaxation and stress relief, particularly in academically demanding environments, excessive consumption may interfere with important aspects of student life.

Academic engagement is a key area that is influenced by digital entertainment consumption. Academic engagement is the level of interest, participation, and commitment that students bring to their educational activities. It entails attending lectures, completing assignments, participating in discussions, and practicing consistent study habits. When students devote a significant amount of their time to digital entertainment, they may have less time and attention available for academic tasks, affecting their overall academic performance.

In addition to academic engagement, digital entertainment consumption influences students' lifestyle balance. Lifestyle balance is the ability to strike a healthy balance between various aspects of life, such as academic responsibilities, physical health, social interactions, and personal well-being. Excessive use of digital entertainment platforms can disrupt this balance

by influencing sleep patterns, decreasing physical activity, and limiting face-to-face social interactions. Long periods of screen time, particularly late at night, have been linked to poor sleep quality, which has an impact on cognitive functioning and productivity.

However, it is critical to understand that digital entertainment is not inherently negative. When used in moderation, it can provide a variety of benefits, including relaxation, stress relief, and even opportunities for learning and creativity. Educational videos, informative podcasts, and interactive content can help to expand knowledge and provide new learning experiences. As a result, the impact of digital entertainment is heavily influenced by the frequency and extent with which it is consumed.

The growing concern among educators and researchers about the potential negative effects of excessive digital consumption emphasizes the need for a better understanding of the phenomenon. While several studies have looked at the effects of digital media on academic performance or lifestyle factors separately, there is little research that investigates both academic engagement and lifestyle balance in relation to digital entertainment consumption. This gap highlights the importance of conducting a thorough analysis that takes into account all aspects of student life.

The primary goal of this research is to examine digital entertainment consumption patterns among university students and its impact on academic engagement and lifestyle balance. The study specifically seeks to determine the extent to which time spent on digital entertainment affects study habits, class participation, sleep patterns, and overall well-being. Furthermore, the study wants to know if moderate consumption can have positive effects, such as improved relaxation and stress reduction.

LITERATURE REVIEW

The rapid development of digital technology has had a significant impact on entertainment consumption patterns, particularly among university students. With the rise of smartphones, streaming platforms, and social media, researchers are increasingly interested in understanding how digital entertainment influences academic engagement and lifestyle balance.

Several studies have investigated the link between digital media use and academic performance. According to Anderson and Jiang (2018), students' increased screen time is associated with decreased academic focus and achievement. Their findings indicate that excessive use of digital platforms can distract students from their academic responsibilities. Similarly, Rosen et al. (2019) discovered that students who frequently multitask with digital devices during study hours have lower retention rates and academic performance.

In addition to academic engagement, researchers have investigated the impact of digital entertainment on cognitive and behavioral factors. Junco (2012) found that excessive use of social media platforms has a negative impact on students' study time and academic involvement. The study emphasized that spending time on platforms like Facebook reduces time available for educational activities. Furthermore, Kirschner and Karpinski (2010) discovered that students who actively participate in social networking have lower grade point averages than those who use such platforms infrequently.

Digital entertainment has a significant impact on students' lifestyle balance. According to Twenge (2019), adolescents and young adults who spend more time on screens have lower sleep duration and quality. Sleep is essential for cognitive function, and disrupting it can have a negative impact on academic performance. Similarly, Exelmans and Van den Bulck (2016) discovered that using digital devices at night causes delayed sleep onset, leading to fatigue and reduced productivity during the day.

Furthermore, Kuss and Griffiths (2017) investigated online gaming behavior and discovered that excessive gaming can lead to addictive tendencies. Such behaviors not only have an impact on academic performance, but they also decrease physical activity and social interaction, disrupting overall lifestyle balance. In contrast, moderate use of digital entertainment has been shown to provide psychological benefits. Przybylski (2014) discovered that limited use of digital games can improve mood, reduce stress, and increase overall well-being.

Another key point raised in the literature is the role of digital platforms in providing educational benefits. Some researchers argue that when used properly, digital media can help students learn. Educational videos, online tutorials, and interactive applications, for example, can help students better understand and engage with academic subjects. This suggests that the impact of digital entertainment is not entirely negative, but rather depends on the nature and duration of use.

Despite the extensive body of research, the majority of studies have focused solely on academic performance or lifestyle factors. There is little research into the combined impact of digital entertainment consumption on academic engagement and lifestyle balance among university students. Furthermore, many studies do not distinguish between types of digital entertainment, such as social media, streaming, and gaming, which may have different effects on students.

Research Gap: Although previous studies have provided useful insights into the effects of digital entertainment, there is a lack of comprehensive research that examines its impact on both academic engagement and lifestyle balance. Furthermore, little effort has been made to understand how different patterns of digital consumption influence these outcomes. As a result, the purpose of this study is to close this gap by conducting an integrated analysis of digital entertainment consumption patterns and their effects on academic and lifestyle behaviors among university students.

DATA COLLECTION & METHODOLOGY

3.1 Data Collection and Sample

Data were collected through a structured online survey administered via Google Forms in April 2026. The data was collected over a period of 2 weeks. The instrument comprised 38 items covering demographic background, daily screen time, primary entertainment platforms, gaming behaviour, academic self-assessment, and wellbeing. Responses were measured on a five-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree) for attitudinal items, alongside categorical responses for demographics and screen time. The final usable sample consisted of n = 150 respondents after removing duplicate entries identified during cleaning. The study was restricted to self-reported university-affiliated respondents, predominantly postgraduate and undergraduate students, with additional representation from PhD scholars and housewives. Data was collected anonymously to ensure confidentiality and reliability. Respondents in the sample come from a variety of academic backgrounds, including Business/Commerce, Engineering/Technology, Science, and Arts/Social Sciences.

3.2 Feature Engineering

Three composite variables were constructed to capture latent behavioral constructs:

Variable	Construction Logic	Scale
Entertainment Addiction Score (EAS)	Mean of Likert items measuring compulsive use, inability to reduce usage, disrupted routines, and pre-sleep consumption behavior	1–5 (continuous)
Academic Performance Index (API)	Mean of items capturing self-reported academic focus, assignment completion, class engagement, and time management	1–5 (continuous)
Wellbeing Score	Mean of items covering sleep consistency, physical energy, stress levels, and sense of balance	1–5 (continuous)

Gaming Intensity	Mean of items related to gaming frequency, absorption, overrun behavior, and gaming as a primary hobby	1–5 (continuous)
Risk Category	EAS-based categorization: Low (≤ 2.5), Moderate (2.5–3.5), High (> 3.5)	3-level ordinal

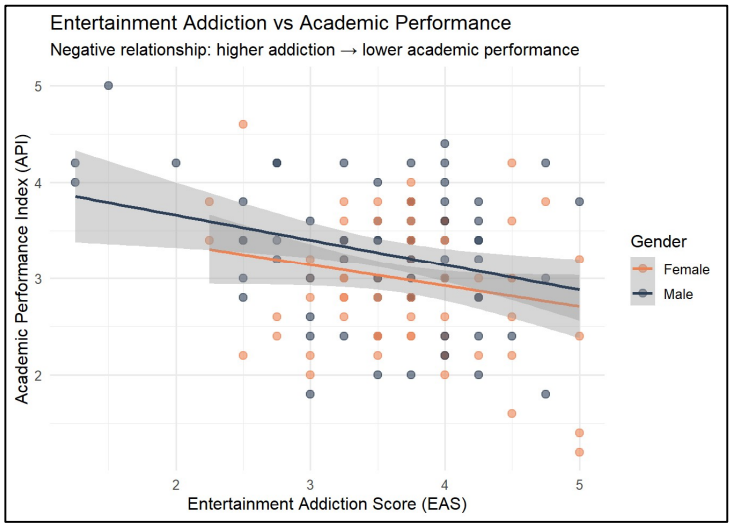
3.3 Analytical Framework

The analysis was structured into five progressive stages implemented in R (version 4.3) using tidyverse, ggplot2, factoextra, and car packages:

Stage	Method	Purpose
EDA	Univariate and bivariate visualizations (histograms, boxplots, scatter plots, stacked bars)	Describe distributions and identify initial patterns in the data
Correlation	Pearson correlation heatmap on numeric variables	Identify linear relationships among key constructs
Regression	OLS Models 1 and 2 (hierarchical entry)	Estimate predictors of API and assess explained variance
PCA	Principal Component Analysis on standardized Likert matrix	Reduce dimensionality and identify latent behavioral factors
Clustering	K-means (k = 3, 25 random starts, 100 iterations); Silhouette analysis	Segment respondents into distinct behavioral archetypes
ANOVA	One-way ANOVA + Tukey HSD; Levene's test for homogeneity	Test mean differences in API across risk categories

RESULTS

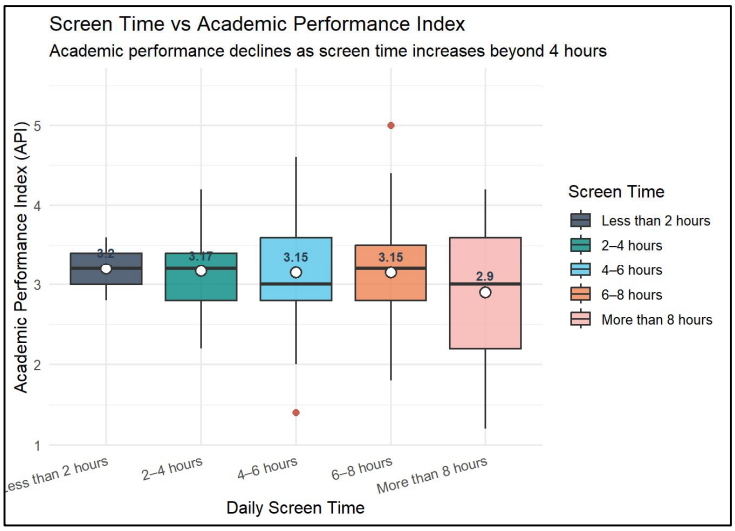
4.1 EAS vs Academic Performance Index (KEY relationship)



Interpretation:

The plot shows a clear negative relationship between Entertainment Addiction Score (EAS) and Academic Performance Index (API), indicating that higher addiction is associated with lower academic performance. The downward trend is consistent across both genders, though slight variations exist in individual data points. Most respondents with high EAS scores tend to have API values clustered on the lower side. This suggests that increased engagement in entertainment activities may adversely impact academic outcomes.

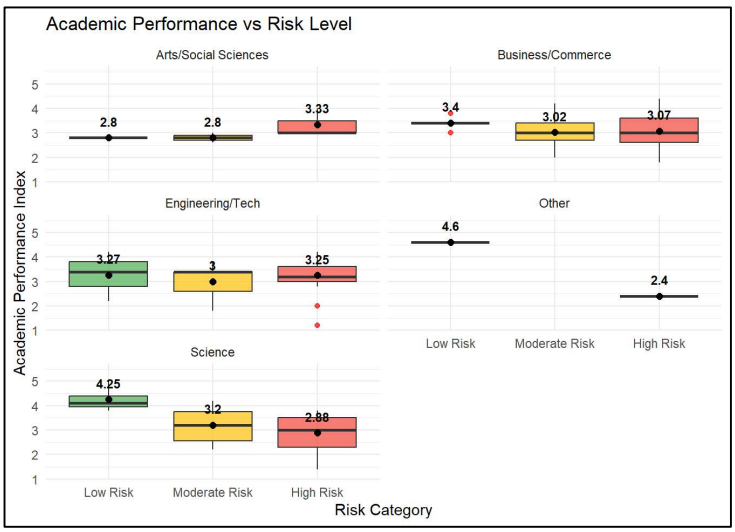
4.2 Screen Time vs Academic Performance Index



Interpretation:

The boxplot shows that academic performance (API) tends to decline as screen time increases, particularly beyond 4 hours per day. Respondents with lower screen time (less than 2–4 hours) have relatively higher and more consistent API scores. As screen time rises to 6–8 hours and above, the median API decreases and variability increases, indicating less stable performance. The lowest scores are more common in the “More than 8 hours” group. Overall, higher screen time is associated with reduced academic performance.

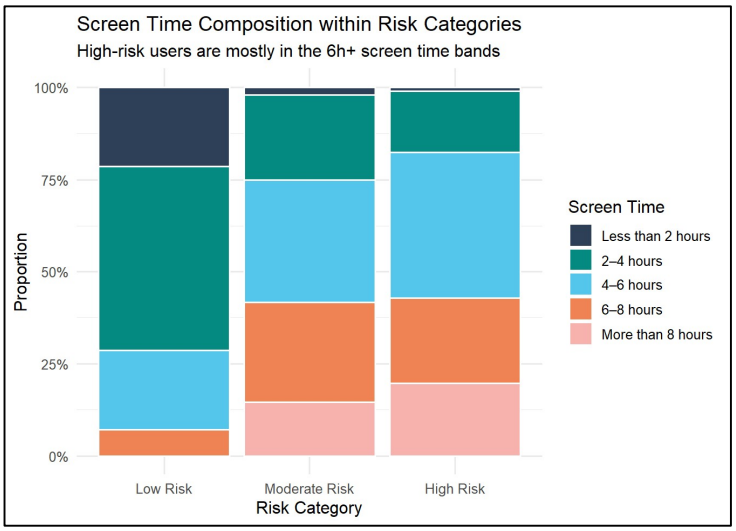
4.3 API faceted by Field of Study and Risk Category



Interpretation:

The faceted plots show that academic performance generally declines as risk level increases across most fields of study. Students in the Low Risk category consistently have higher API scores, while Moderate and High Risk groups show lower and more variable performance. This trend is especially clear in Science and Engineering/Tech, where the drop from low to high risk is more pronounced. In Business/Commerce and Arts, the decline is present but slightly less consistent. Overall, higher entertainment risk is associated with poorer academic outcomes across disciplines.

4.4 Screen time composition within risk groups



Interpretation:

The stacked bar chart shows that higher risk categories are associated with greater proportions of users in the higher screen time bands. High Risk individuals are predominantly concentrated in the 6–8 hours and more than 8 hours categories, indicating heavy screen usage. In contrast, Low Risk respondents are more represented in the lower screen time ranges, especially 2–4 hours. Moderate Risk users display a more balanced distribution across categories. Overall, increased screen time is clearly linked to higher entertainment addiction risk levels.

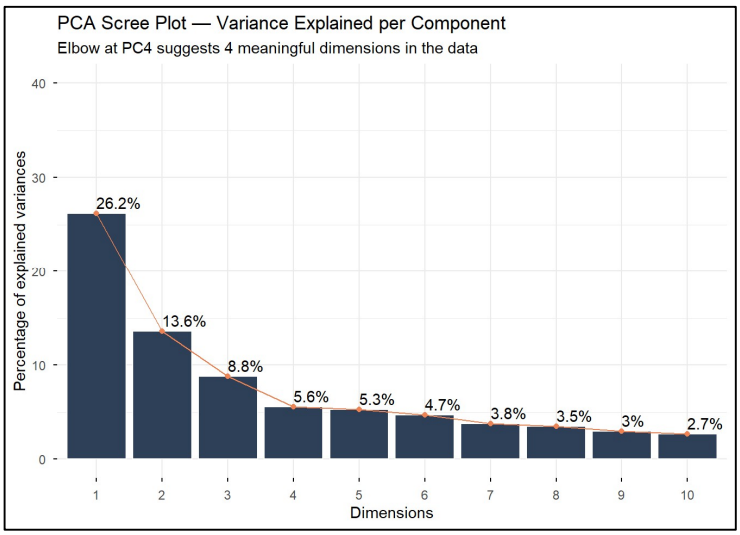
4.5 PRINCIPAL COMPONENT ANALYSIS (PCA)

```
Table: PCA Variance Explained (first 10 PCs)

|PC   | Variance_Pct| Cumulative|
|:----|------------:|----------:|
|PC1  |        26.18|      26.18|
|PC2  |        13.62|      39.80|
|PC3  |         8.81|      48.62|
|PC4  |         5.59|      54.20|
|PC5  |         5.29|      59.49|
|PC6  |         4.70|      64.20|
|PC7  |         3.81|      68.00|
|PC8  |         3.51|      71.51|
|PC9  |         2.98|      74.49|
|PC10 |         2.68|      77.17|

▶ Code

  → PC1 + PC2 explain 39.8% of total variance

▶ Code

  → First 4 PCs together explain 54.2% of total variance
```

Interpretation:

The PCA results show that the first principal component (PC1) explains the largest share of variance (26.18%), followed by PC2 (13.62%), together accounting for 39.8% of the total variability in the data. This indicates that a significant portion of the underlying patterns can be captured using just two dimensions. The first four components explain 54.2% of the variance, suggesting moderate dimensionality reduction while retaining key information. Beyond PC4, each component contributes relatively smaller variance, indicating diminishing importance. Overall, PCA effectively reduces complexity while preserving the main structure of the dataset.

4.6 Scree plot



Interpretation:

The scree plot shows that variance explained drops sharply from PC1 to PC4, after which the decline becomes gradual, forming an elbow at PC4. This indicates that the first four principal components capture the most meaningful structure in the data, while additional components contribute relatively little. PC1 and PC2 explain the largest portions, with diminishing returns from subsequent components. The flattening of the curve beyond PC4 suggests limited added value in retaining more dimensions. Overall, the plot supports selecting four components for efficient dimensionality reduction.

4.7 Top loadings on PC1

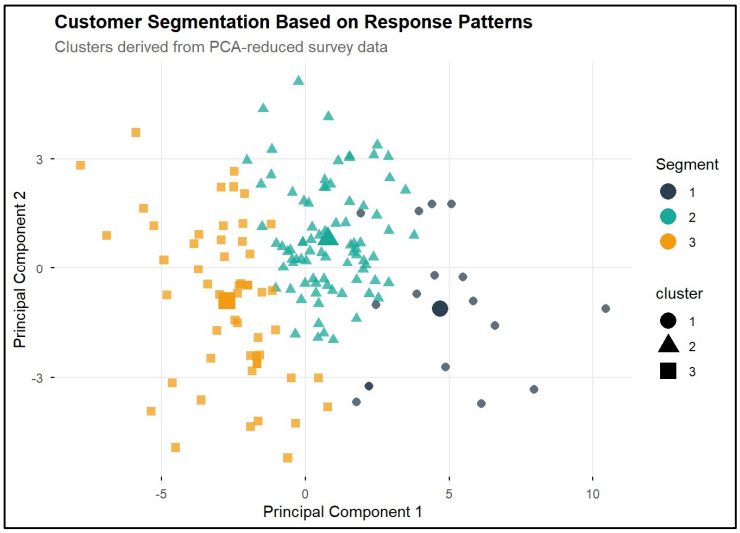
```
Top 10 variable loadings on PC1 (largest absolute values):

▶ Code

       game_overrun       daily_routine       game_absorbed     relax_via_video
              0.254               0.250               0.243               0.242
   presleep_content multitask_entertain    game_major_hobby        scroll_short
              0.236               0.231               0.226               0.224
       game_improve      scroll_overrun
              0.223               0.218
```

Interpretation:

PC1 captures general 'digital entertainment immersion' ; items about binge-watching, scrolling, gaming, and boredom-driven usage load together. PC2 likely captures 'academic self-regulation' vs. 'entertainment escapism'. High-risk respondents cluster distinctly from low-risk in PC space, confirming that entertainment behavior forms coherent behavioral profiles.

4.8 K-MEANS CLUSTERING

Cluster visualization on PCA space



Interpretation:

The clustering plot shows three distinct respondent segments based on their response patterns in the PCA-reduced space. Cluster 1 is clearly separated on the positive side of PC1, indicating a unique behavioral profile compared to others. Cluster 2 is centrally located, suggesting moderate or balanced response patterns. Cluster 3 is positioned on the negative side of PC1,

reflecting a contrasting set of behaviors. Overall, the clusters are reasonably well-separated, supporting meaningful segmentation of respondents into distinct groups.

4.9 MULTIPLE LINEAR REGRESSION

```
▶ Code

Model 1 R²: 0.075 (explains 7.5% of API variance)

▶ Code

Model 2 R²: 0.203 (explains 20.3% of API variance)
```

Interpretation:

The regression results show that in Model 1, entertainment addiction has a significant negative impact on academic performance, while screen time is not a significant predictor, explaining only 7.5% of the variance. In Model 2, the overall explanatory power improves ($R^2 = 20.3\%$), but entertainment addiction and screen time lose significance after adding other variables. Wellbeing score emerges as a strong positive and significant predictor of academic performance. Other factors like gaming, age, and field of study are not significant, while gender shows a weak effect. Overall, wellbeing plays a more critical role in predicting academic outcomes than direct entertainment usage.

4.10 ONE-WAY ANOVA — Does risk category SIGNIFICANTLY affect API?

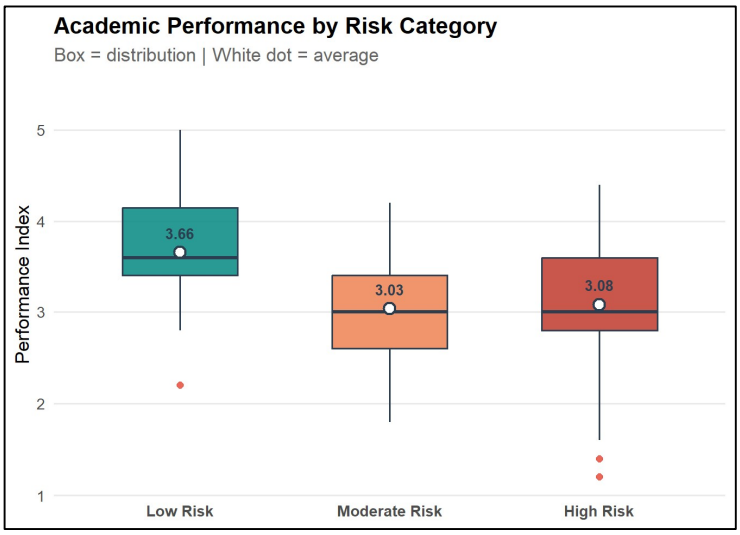
```
▶ Code

  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = academic_performance_index ~ risk_category, data = df)

$risk_category
                              diff        lwr        upr     p adj
Moderate Risk-Low Risk  -0.62380952 -1.0889485 -0.1586705 0.0051471
High Risk-Low Risk      -0.57802198 -1.0176461 -0.1383979 0.0062567
High Risk-Moderate Risk  0.04578755 -0.2273854  0.3189605 0.9169361
```

Interpretation:

The ANOVA results show a statistically significant difference in academic performance across risk categories ($p = 0.005$), indicating that at least one group differs in mean API. The Tukey post-hoc test reveals that both Moderate Risk and High Risk groups have significantly lower

academic performance compared to the Low Risk group. However, there is no significant difference between Moderate and High Risk groups. This suggests that moving from low to higher risk levels is associated with a decline in academic performance, but differences within higher risk groups are minimal.

ANOVA result visualization



Interpretation:

The boxplot shows that Low Risk respondents have the highest average academic performance, while both Moderate and High Risk groups have lower mean API scores. The decline from Low to higher risk levels is clearly visible, supporting the ANOVA results. Moderate and High Risk groups show similar central tendencies, with overlapping distributions. Greater variability is observed in the High Risk group, indicating inconsistent performance. Overall, higher risk levels are associated with lower academic performance.

4.11 Levene's test (homogeneity of variance)

```
Levene's Test for Homogeneity of Variance (center = median)
       Df F value Pr(>F)
group   2  0.3507 0.7048
      150
```

Interpretation:

Levene's test shows that the assumption of equal variances is satisfied ($p = 0.7048 > 0.05$), indicating no significant difference in variance across the risk groups. This means the

variability in academic performance is similar for Low, Moderate, and High Risk categories. As a result, the homogeneity of variance assumption required for ANOVA is met. Therefore, the ANOVA results can be considered reliable and valid.

4.12 Key Statistics Summary

Table: Risk Group Summary Statistics

risk_category	n	mean_EAS	mean_API	mean_sleep	mean_energy
Low Risk	14	2.18	3.66	2.93	4.00
Moderate Risk	48	3.22	3.03	2.83	3.29
High Risk	91	4.09	3.08	2.93	3.38

Interpretation:

The summary statistics show a clear gradient across risk categories, with entertainment addiction increasing from Low (2.18) to High Risk (4.09). Academic performance is highest in the Low Risk group (3.66) and drops in Moderate and High Risk groups, though the difference between the latter two is minimal. Sleep consistency remains relatively similar across groups, while energy levels are highest in Low Risk and lower in higher risk categories. This suggests that higher addiction is associated with reduced academic performance and slightly poorer energy levels. Overall, Low Risk individuals exhibit the most favorable outcomes.

CONCLUSION

This study analyses digital entertainment consumption and its impact on academic engagement and lifestyle balance among university students using a multivariate approach. Results show a high prevalence of risky consumption behaviours, with significant differences in academic performance across risk groups (ANOVA, $p < 0.01$). Correlation and regression analyses indicate a negative relationship between entertainment addiction and academic performance, while wellbeing emerges as a strong positive predictor ($p < 0.001$). The full model explains 20.3% of the variance, suggesting moderate explanatory power. Principal Component Analysis (PCA) captures over 54% of variance within the first four components, supporting dimensionality reduction. K-means clustering identifies distinct behavioral segments, ranging from low-risk high-performing to high-risk low-wellbeing groups. Overall, the findings highlight that behavioral intensity and wellbeing, rather than screen time alone, significantly influence academic outcomes.

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